From Feedback to Insight: Understanding Amazon Reviews through Text
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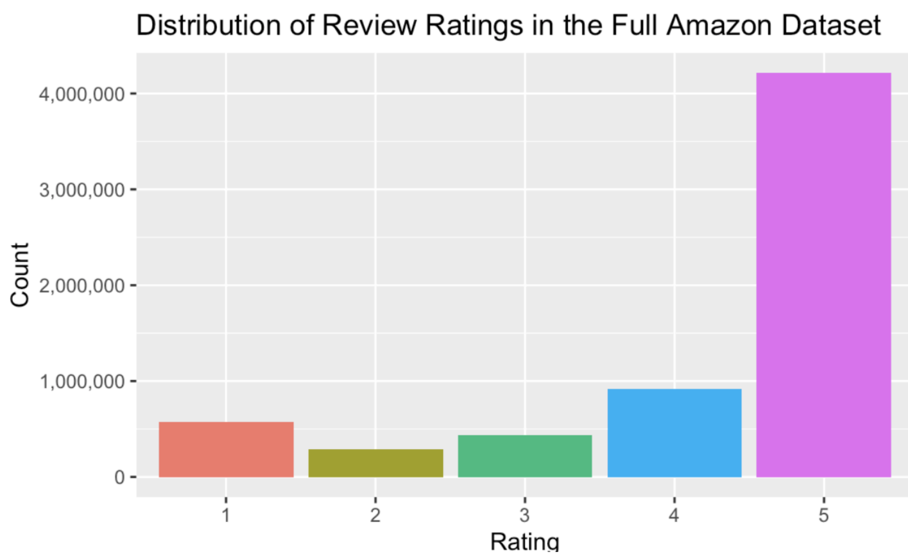
Introduction

Our team analyzed a dataset of 6.4 million Amazon customer product reviews to answer, "What language patterns are most associated with low- versus high-rated reviews?". Our goal was to uncover themes that inform business insights and guide marketing strategies. We preprocessed text data into binary word indicators and trained logistic regression and random forest classification models to predict review ratings. Both models achieved high accuracy and showed positive reviews emphasized praise, while negative reviews emphasized malfunction. This report summarizes our data cleaning, CHTC workflow, model results, and limitations.

Data Cleaning, Text Classification

The Amazon Reviews dataset from Kaggle contained 6.4 million rows, each representing a single customer review. The dataset contained 9 variables, while overall rating (1-5 stars) and reviewText were most important for our analysis. It became clear that the dataset was highly imbalanced, with significantly more high reviews than low.



To clean the data, we extracted reviewText and labeled 1-2 star reviews as low-rated, 4-5 star reviews as high-rated, and removed 3-star reviews. We removed punctuation, tokenized the text, removed blank entries, and filtered out filler words like "the". We then identified the top 100 most common words to use as model features and converted text into binary indicators, assigning 1 if the word was present and 0 if not.

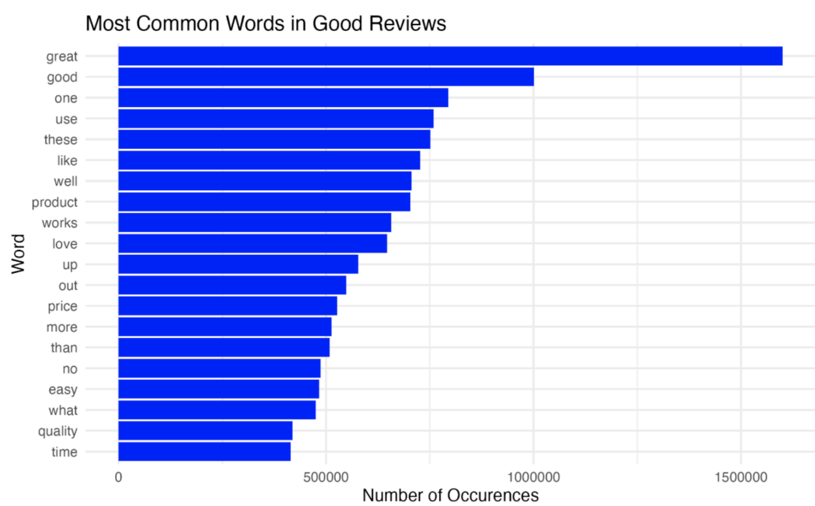
Statistical Computation

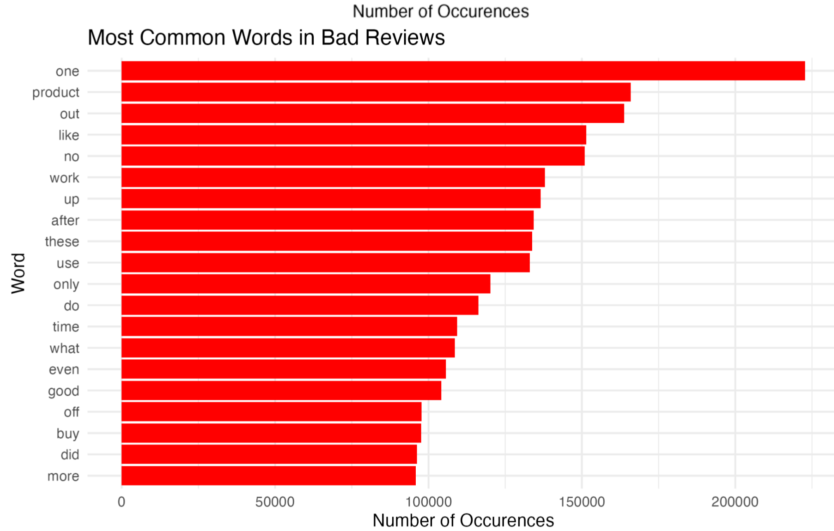
Our computational workflow used CHTC to process the Amazon review text because the dataset was 37 GB. This included splitting the dataset into smaller chunks, processing jobs independently, and then merging.

We split the dataset into chunks of 250,000 lines each assigned to separate CHTC jobs. Each chunk read each JSON review, extracting overall rating and reviewText, skipping unreadable lines, and grouping reviews as low- or high-rated. The script converted text to lowercase, removed punctuation and non-letter characters, tokenized words, removed stop words, and standardized contractions like “doesnt”.

Each job used 1 CPU, 2 GB of memory, and 4 GB of disk, producing two outputs: a partial word-count file with group, word, and count columns and a summary file with complete records, low- and high-rated reviews, and improperly formatted lines. We ran 26 CHTC jobs and each took about 10 minutes.

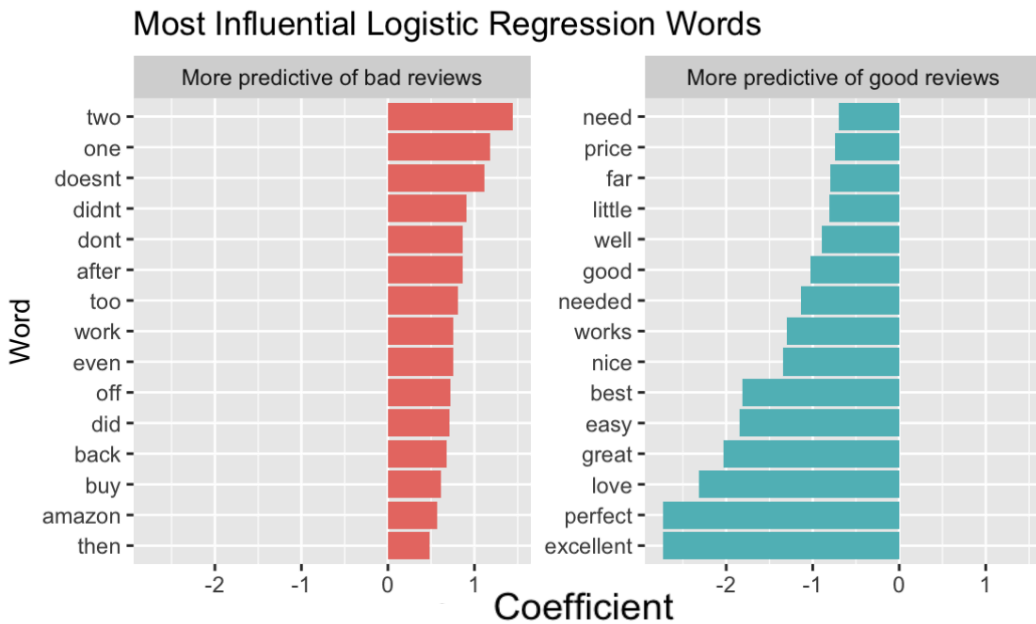
We merged partial word-count outputs by removing duplicate headers, cleaning formatting issues, and summing counts for each rating group and word. This produced final grouped word counts and top-word lists for low and high-rated reviews. Below are the top 20 words associated with low- and high-rated reviews.



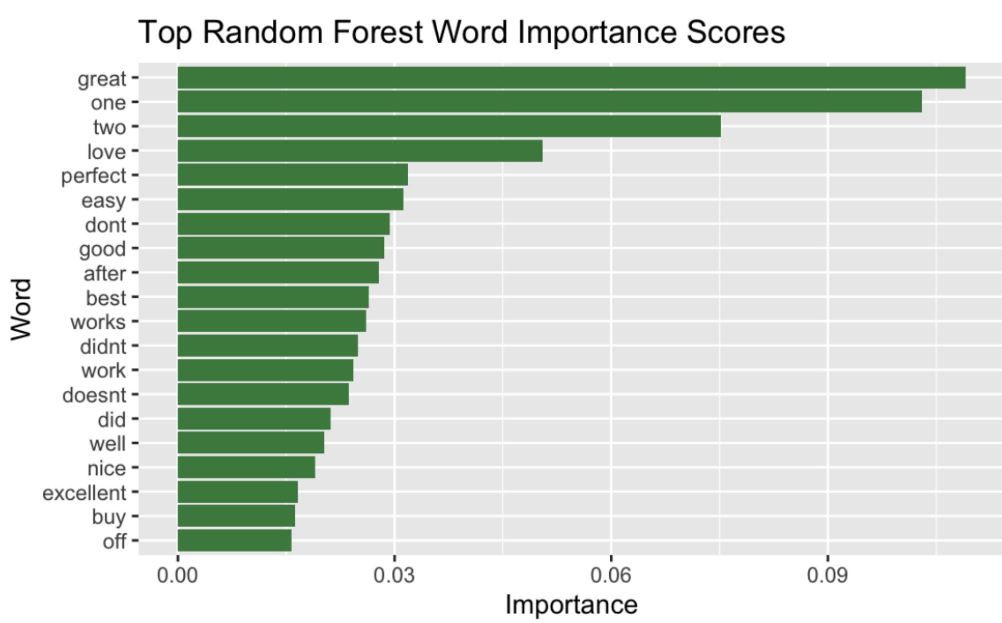


Results and Limitations/Challenges

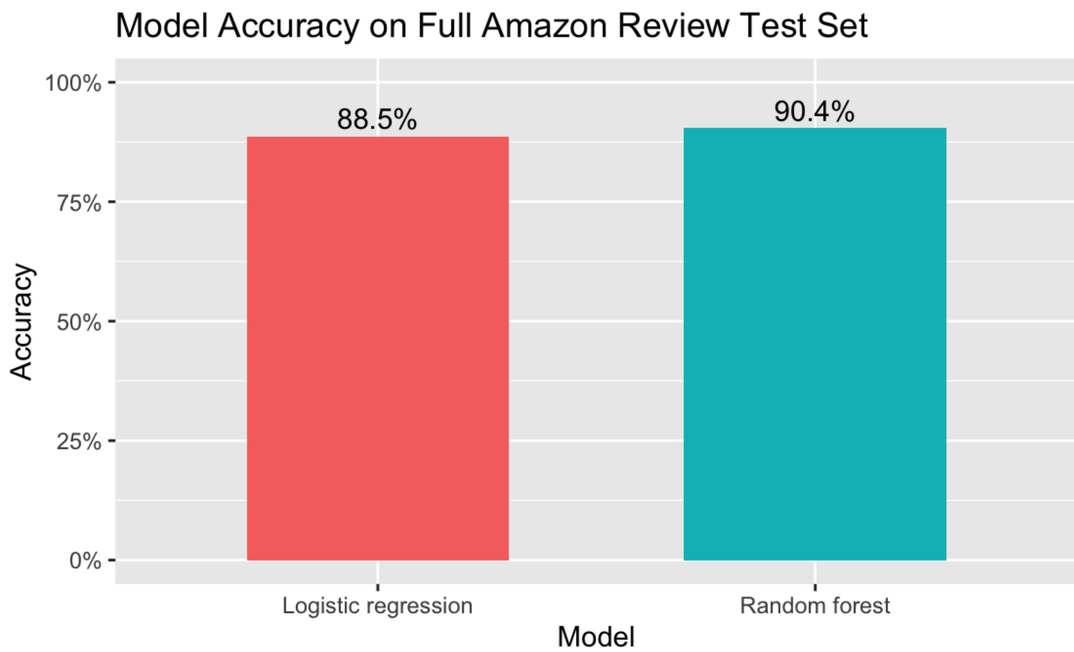
Logistic regression provided direction and strength through coefficients. Words farther from zero had stronger effects, with some pushing predictions toward low-rated reviews and others toward high-rated reviews. Words like “doesnt” suggest dissatisfaction was tied to malfunction, while words like “love” showed satisfaction was tied to emotion.



Random forest provided importance scores, which highlighted words useful for classification but not direction of their effect.



The random forest and logistic regression model achieved 90.4% and 88.5% accuracy respectively.



The models showed that failure-related words identified low-rated reviews, while praise-related words identified high-rated reviews.

Some limitations are the dataset skewed positive, and using single words separated contextual phrases like “doesn't work” or “don’t buy”.

Conclusion

Our project investigated language patterns that distinguish positive and negative Amazon reviews. High-rated reviews were associated with praise, while low-rated reviews were tied to faulty products. Future work should use a balanced dataset and include longer phrases such as “doesn’t work”.

Contributions

Member	Proposal	Coding	Presentation	Report
Phoebe Tseng	.5	.5	1	1
Sophia Taffet	1	1	1	1
Sofia Caceres	1	1	1	.5
Ashvath Madhushankar	1	.5	1	1
Sambhav Mahajan	0	0	1	.5